Mounji Malak

Engineering Student, Data Science · Big Data · Artificial Intelligence

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Portfolio mounjimallak-bit.github.io/Portfolio: projects, code and case studies

PROFILE

Final-year engineering student in data science. I work on anomaly detection in sensor time series: learning what normal operation looks like from data, and testing how far such models can be trusted when labelled events are rare and conditions change.

EDUCATION

ENSA Agadir, National School of Applied Sciences — Agadir, Morocco

State Engineering Degree (Diplôme d'Ingénieur d'État), Data Science, Big Data & Artificial Intelligence — 2022 to 2027 (expected)

5 years at Master's level: 2 years of integrated preparatory classes, then 3 years of engineering cycle. Relevant coursework: Stochastic Processes · AI in Industrial Environments · Data Mining · AI Modeling · Sparsity in Machine Learning · AI Algorithms and Prediction · Big Data

IBN MAJA High School — Scientific Baccalaureate, With Highest Honors (18/20) — 2022

Mohammed VI Foundation Excellence Scholarship, awarded on Baccalaureate results, held 2022 to 2025 (full duration of the award).

EXPERIENCE

Machine Learning Engineer Intern (PFA) — July 2026 to August 2026

Albarray Consulting, anomaly detection for photovoltaic plants — *Agadir, Morocco*

- Built a semi-supervised anomaly detection pipeline for a PV installation: autoencoders trained on healthy operation only, evaluated on a strictly chronological split against staged shading faults.
- Fused 5 dense and 5 LSTM autoencoders by rank (test PR-AUC 0.78, 3/3 fault episodes); compared them with kNN and Mahalanobis baselines, quantified uncertainty (exact intervals, day-level bootstrap) and measured seasonal false alarms on 10.5 fault-free months (Scikit-learn, TensorFlow/Keras).
- Built a streaming prototype: Kafka ingestion, Python scoring consumer, TimescaleDB storage, alarm topic and a Grafana dashboard provisioned as code; Docker Compose; automated tests showing streaming and offline scores match.

Code: github.com/mounjimallak-bit/pfa-photovoltaique

Data Analyst / Data Scientist Intern — July 2025

HCP, High Commission for Planning — *Morocco*

- Structured and harmonised public statistical tables into datasets ready for analysis and institutional decision making.
- Applied trend extrapolation models (Scikit-learn, linear regression) to socioeconomic indicator series.
- Built a decision support chatbot (Google Gemini API, Streamlit) letting nontechnical users query indicators in natural language.
- Automated recurring analytical reporting (PDF: statistics, ranking tables, charts), replacing a manual spreadsheet process.

PROJECTS

IoT–Edge–Cloud pipeline: next-day Delta Air Lines stock return prediction — 2026, in progress

Group course project, ENSA Agadir

- Question: do real-time operational disruptions (flight delays and cancellations, airport weather) help predict the next-day return of Delta Air Lines stock?
- Sources: aircraft positions (OpenSky ADS-B) and airport weather stations (IEM) as IoT streams, BTS on-time flight records as enterprise data, daily market data. Pipeline: MQTT → Edge filtering and validation → Kafka → Spark (streaming and batch) → MinIO (Bronze/Silver/Gold) → Streamlit.
- Planned evaluation: chronological split, Random Forest compared with a zero-return baseline and with the same model without flight features.

Axiora Data, Socioeconomic Analytics Platform (Morocco) — 2025

Personal project extending my work at HCP: axiora-data.streamlit.app

- 93 hand-transcribed HCP tables for Souss-Massa with automated consistency checks (totals, table structure, map joins), and a descriptive analysis of commune-level poverty change 2004–2014, compared with population-weighted territory rates, reproducible from the repository.
- Streamlit application with regional and commune choropleth maps (GeoPandas, Plotly), automated PDF reporting and a natural language query layer.

Other coursework projects: Carrefour sales analysis (Python ETL, Power BI star schema, DAX), team project, 2025 · Real estate rental platform (Flask, Oracle PL/SQL triggers and stored procedures), 2024.

TECHNICAL SKILLS

Machine Learning	Scikit-learn · TensorFlow / Keras · Anomaly detection (autoencoders, LSTM autoencoders, Isolation Forest, kNN and Mahalanobis baselines) · Classification · Regression · Clustering
Statistics	Bootstrap and interval estimation · Non-parametric tests · Time-series evaluation
Big Data & Streaming	Apache Kafka · Spark (SQL, Streaming, MLlib) · Hadoop · HDFS · Hive
Deployment	Docker and Docker Compose · PostgreSQL / TimescaleDB · Grafana · Git / GitHub
Programming	Python (pandas, NumPy, SciPy) · SQL · R · C · LaTeX
Data & Tools	Plotly · Matplotlib · GeoPandas · Streamlit · Power BI · MySQL · Oracle SQL · MongoDB · Jupyter

LANGUAGES & ACTIVITIES

Languages	Arabic (Native) · French (C1) · English (C1)
Tutoring	Tutored 1st to 3rd year students at ENSA Agadir over several years
Volunteering	Amal Elkhayr association: renovation of a rural primary school near Agadir